GoTourGo: Smart Travel Companion

Functional MVP Development Proposal

1. Project Overview

GoTourGo is a mobile travel companion designed to help users explore destinations, discover nearby places, plan visits, access travel information, and stay connected with friends while travelling.

The objective of this project is to develop a functional Android MVP that demonstrates the core concepts and major features of the proposed GoTourGo application.

The implementation will prioritize working features, a user-friendly interface, and cost-effective technologies. Some features will use live services, while others will use predefined data or simplified demonstrations where appropriate for the university project.

2. Proposed Technology Stack

- **Mobile Application:** React Native with Expo
- **Database and Backend Services:** Supabase
- **Maps:** MapLibre with a compatible map data provider
- **Location Services:** Device GPS
- **AI Features:** Gemini API, where applicable
- **Local Storage:** For saved data and selected offline functionality
- **Android Build:** Installable APK

The application will use free-tier services and open-source tools wherever practical. Third-party API usage, service quotas, and provider restrictions will determine which integrations can be delivered without additional costs.

3. Development Phases and Features

Phase 1: Intelligent Mapping and Location-Based Discovery

This phase will provide the application's primary mapping and location functionality.

Features:

- Interactive map with destination markers.
- Display the user's current location using GPS.
- Browse selected tourist attractions and points of interest.
- Display basic information about supported destinations.
- Select a destination and view its location on the map.
- Basic location-based alerts and proximity notifications for supported destinations.
- Online route calculation through a compatible routing service.

- Initial offline-map functionality for a selected, downloaded map region, subject to provider compatibility and technical testing.

Destination information may combine predefined records with live map data where appropriate.

Phase 2: Safety, Connectivity and Social Interaction

This phase will combine features that involve user accounts, location sharing, and interaction.

Features:

- User registration and login.
- Basic user profiles.
- Add or connect with friends within the application.
- Share location with an accepted friend, subject to the application's permissions and connectivity.
- Display participating friends on the map when their location is available.
- Basic SOS interface with a mechanism to share the user's location with a designated contact.
- Community posts or destination reviews.
- Basic destination ratings and user feedback.

Real-time tracking will depend on an active internet connection and the relevant location permissions. Emergency features will be implemented as a basic demonstration and will not be represented as a verified emergency-response service.

Phase 3: AI-Powered Travel Assistance and Visual Search

This phase will introduce a simplified AI-assisted travel experience.

Features:

- Basic AI travel assistant powered by the Gemini API.
- Answers to supported travel-related questions.
- Travel guidance based on the information provided to the AI.
- A simple visual-search demonstration using an appropriate image input workflow.
- Display relevant destination information when supported by the available data.

The implementation will use predefined destination information and controlled prompts where suitable. Advanced image recognition, comprehensive destination identification, and a full retrieval-augmented generation system are outside the initial MVP scope.

AI functionality will depend on API availability, applicable quotas, and provider requirements.

Phase 4: Trip Checklist and Destination Management

This phase will allow users to organize the places they want to visit.

Features:

- Search for a destination or select a location on the map.
- Add a destination to the user's trip list.
- Assign a status such as To Do, In Progress, or Visited.
- Add optional notes or a planned visit date.
- Update or remove saved destinations.
- View saved destinations and their statuses.
- Save trip-list information to the database, with local access to previously saved information where implemented.

The trip planner will function as a simple destination checklist rather than an AI-generated itinerary system.

Phase 5: Gamification and Travel Rewards

This phase will introduce basic progress tracking and recognition.

Features:

- Track destinations marked as visited.
- Display the number of places visited.
- Calculate or display travel distance where suitable location data is available.
- Award simple badges for defined milestones.
- Display a basic leaderboard based on the supported progress metrics.

The leaderboard and rewards will use the data available to the application. Advanced anti-cheating mechanisms and complex reward systems are not included.

Phase 6: Immersive Content and Entertainment

This phase will provide a simplified implementation of the proposed immersive and entertainment features.

Features:

- A dedicated VR/AR section within the application.
- A Coming Soon screen to represent the planned immersive experience.
- A music section containing approximately 10 to 20 predefined tracks seeded into the application or its database.
- Basic music playback controls, including play, pause, and track selection.

Actual VR/AR functionality, video previews, interactive 3D environments, and advanced immersive experiences will not be implemented in this MVP.

Phase 7: Destination Information and Travel Dashboard

This phase will bring the application's main information and progress features together.

Features:

- A home screen with shortcuts to the main application sections.
- Destination information cards.
- Selected nearby services or infrastructure markers where supported by the map data.
- Basic weather information through a suitable service, if accessible within the available quota.
- A user dashboard showing saved trips, visited destinations, and achievement progress.
- Basic loading, empty, and error states to make the application usable.

Some information may be predefined, while other information may be retrieved from available APIs.

4. Database and Backend

Supabase will be used for the application's core data storage and backend functionality.

Depending on the implemented features, the database may contain:

- User profiles.
- Destination records.
- Saved destinations and trip-list statuses.
- Friend connections and location-sharing information.
- Community posts and destination reviews.
- Travel achievements and leaderboard records.
- Seeded music track records.
- Other basic records required by the implemented features.

The database structure will remain focused on the MVP requirements rather than a large-scale production architecture.

5. Offline Functionality

The application will support a limited offline experience for selected features.

The intended scope includes:

- Access to a previously downloaded map region, if supported by the chosen map-data provider.
- Continued GPS location updates when the device can obtain a GPS signal.
- Display of a previously calculated and saved route.
- Access to locally saved trip-list information where implemented.

A compatible map-data provider and an offline-capable map implementation will be required. Offline maps and routing will be tested early because their feasibility depends on the selected providers and their technical restrictions.

Full offline turn-by-turn navigation, automatic offline rerouting, and complete offline synchronization across all application features are not included in the initial scope.

6. UI and User Experience

The application will have a consistent mobile interface designed for Android.

The UI will include:

- Main dashboard.
- Interactive map screen.
- Destination details.
- Trip checklist.
- AI assistant.
- Friends and safety section.
- Community and reviews.
- Achievements and leaderboard.
- Entertainment section.
- User profile and settings.

The interface will prioritize straightforward navigation and clear presentation of the implemented features.

7. Testing and Delivery

The project will include:

- Functional testing of implemented features.
- Basic testing of database interactions and API integrations.
- Testing of location permissions and GPS behavior.
- Testing of the selected offline-map functionality on a compatible Android device.
- Fixes for major issues identified during the agreed testing process.
- A final installable Android APK.

Primary deliverable: A working Android APK demonstrating the agreed GoTourGo MVP features.

The project is intended as a FYP prototype, not a fully production-ready commercial application.

8. Scope and Implementation Notes

To keep the project cost-effective, the implementation will prioritize demonstrating the main concepts through a combination of working functionality, predefined data, and simplified feature demonstrations.

The following limitations apply:

- External APIs will be used only where practical and accessible within their applicable quotas and terms.
- Features requiring paid services, specialized infrastructure, or extensive development may be simplified or demonstrated with predefined data.
- Advanced AR/VR, production-grade emergency services, comprehensive offline routing, and complex real-time infrastructure are outside the initial MVP scope.

- Free-tier availability and quotas are controlled by third-party providers and cannot be guaranteed for unlimited use.
- Any material change to the agreed scope will be discussed before additional work is undertaken.

9. Final Deliverable

The goal is to provide a functional demonstration of GoTourGo's main concepts while keeping implementation complexity and external service costs to a minimum.

The final application will demonstrate the agreed features in a usable Android build.